

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30-06-2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Technical Architecture:

Web-Based Nutrition Assistant Architecture

Reference:

- **MongoDB MERN Stack Architecture**
<https://www.mongodb.com/mern-stack>

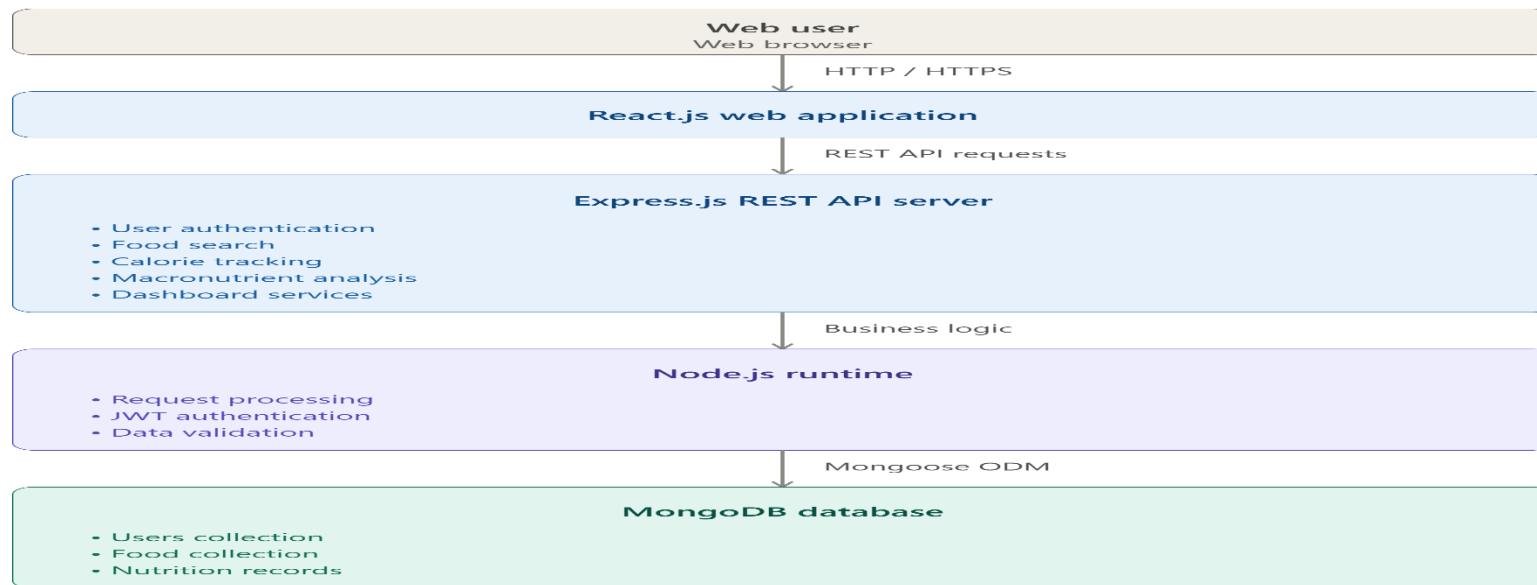


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web-based user interface for registration, login, dashboard, food search, and nutrition tracking	React.js, Tailwind CSS, HTML5, CSS3, JavaScript, Vite
2	Application Logic – Authentication	User registration, login, JWT authentication, session management	Node.js, Express.js, JWT, bcrypt.js
3	Application Logic – Nutrition Management	Food search, calorie tracking, macronutrient analysis, nutrition calculations	Node.js, Express.js
4	Application Logic – Dashboard	Generates daily nutrition summary and dashboard statistics	Node.js, Express.js
5	Database	Stores user accounts, food details, nutrition records	MongoDB, Mongoose ODM
6	Cloud Database	Cloud-hosted NoSQL database	MongoDB Atlas
7	File Storage	Stores application assets and static resources	Local File System / Public Assets
8	External API – Nutrition Data	Retrieves food and nutritional information (<i>if used</i>)	Nutrition API (e.g., USDA FoodData Central / Edamam API) (<i>Otherwise: Not Applicable</i>)
9	REST API Communication	Communication between frontend and backend	Express.js REST APIs using Axios/Fetch
10	Authentication & Security	Secure user authentication and authorization	JWT, bcrypt.js
11	Infrastructure (Deployment)	Frontend deployed on Vercel, Backend deployed on Render, Database hosted on MongoDB Atlas	Vercel, Render, MongoDB Atlas

Table-2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Frameworks and libraries used for frontend and backend development	React.js, Express.js, Node.js, Tailwind CSS, Mongoose, Vite
2	Security Implementations	Password hashing, JWT authentication, input validation, environment variables	bcrypt.js, JWT, dotenv, Express Middleware
3	Scalable Architecture	Three-tier architecture separating presentation, business logic, and database	React.js + Express.js + MongoDB (MERN Stack)
4	Availability	Cloud deployment ensures high availability and accessibility	Vercel, Render, MongoDB Atlas
5	Performance	Optimized React rendering, REST APIs, efficient MongoDB queries, responsive UI	React.js, Express.js, MongoDB, Vite

References:

- **MongoDB Official Documentation**
<https://www.mongodb.com/docs/>
- **React Official Documentation**
<https://react.dev/>
- **Express.js Official Documentation**
<https://expressjs.com/>
- **Node.js Official Documentation**
<https://nodejs.org/docs/latest/api/>